RF Receiver

CN101
CON-SMA-EDGE-S

Through Hole for Antenna
CN102
ULO-PH409-1*2SF11

VPP
VPP=DC7.5V-40V

D101 PME66010AESBYL
D102 PME66010AESBYL
D105 PME66010AESBYL
D106 PME66010AESBYL

C105 1000p/50(COG)
C106 0.1u/100(X7R)
C107 10u/50(X5R)
D104 PWR3Z39VT1G

TP102 TP1.0
TP104 TP1.0
GND

Through Hole for VPP/GND
CN107
ULO-PH409-1*2SF11

R103 27mF(1/4W)
PWR_FLAG
TP103 TP1.0

U101
TS30042-M000QFNR
VCC(abs_max)=42V
BST VSW VSW VSW VSW VCC VCC VCC EN/SYNC FB PG PAD PGND PGND NC NC GND

C101 0.022u/50(X7R)
L101 NR5040T4R7N
D103 S210FA

C102 10u/50(X5R)
C103 0.1u/50(X7R)
TP105 TP1.0

Vout
= 0.9 * (1 + 82k / 18k)
= 5.0 V

C104 0.1u/50(X7R)
C108 22u/16(X5R)
C109 22u/16(X5R)
C110 10u/50(X7R)
R101 82kF
R102 18kF
R104 10kF
PWRGD_5V
TP114 TP1.0

FB101 BLM18PG330SN1D
+5V
TP101 TP1.0

Through Hole for +5V/GND
CN105
ULO-PH409-1*2SF11

Current Monitor

+5V
R114 OPEN
R115 0
R116 10kF
R117 0

U103
PAC1720-1-AIA-TR
SENSE1+ SENSE1- SENSE2+ SENSE2- VDD ADDR_SEL SMCLK SMDATA GND EP ALERT

C116 0.1u/50(X7R)

SCL
SDA
ALERT
TP109 TP1.0

Connect to Transmitter (For Debug)
VEXT TP106
SDA TP107
SCL TP108
TP110
CN103
SM04B-SRSS-TB(LF)(SN)
FG

LiPo Charger

+5V
C118 4.7u/16(X5R)
C119 0.1u/50(X7R)
R119 OPEN
R120 100kF
R126 100kF

U104
MCP73871T-1CCI_ML
IN IN VPCC PG STAT2 STAT1/LBO SEL PROG2 TE CE OUT OUT VBAT VBAT VBAT_SENSE THERM PROG1 PROG3 EP VSS VSS

VCC
C117 4.7u/16(X5R)
C120 4.7u/16(X5R)
GND

R105 OPEN
R106 OPEN
For Battery Monitor
Connect to ADC
CN109
ULO-PH409-1*2SF11

R118 NCP15XH103F03RC
Connect to LiPo
CN104
S2B-PH-SM4-TB(LF)(SN)

R112 SHORT_JUMPER
R123 100kF
R124 10kF
!!!TEMPORARY VALUE!!!
CN110
ULO-PH409-1*2SF11
Through Hole for
External Resistor Setting

PROG1: Fast charge current regulation setting: IREG(mA) = 1000(V)/RPROG1(kohm)
PROG3: Charge terminate current: ITERM(mA) = 1000(V)/RPROG3(kohm)

SEL=H: AC-DC Adaptor
PROG2=L: USB100mA
TE=L: Timer Enable

CHG_EN
CHG_STAT1
CHG_STAT2
CHG_PG
CN112
ULO-PH409-1*4SF11
Through Hole for
Control & Check Chager condition

DC-DC Converter (Step-Up/Down)

VOUT = 0.496*(1+820k/91k) = 4.965V
VOUT = 0.496*(1+820k/143k) = 3.340V

3.3V/5.0V Output Voltage Selector
SW101
CUS-12TB
VD101
CG0402MLC-3.3LG

L102
DFE252012P-3R3M=P2

U102
MP2155GQ-Z
SW1 SW2 VIN VCC VOUT MODE/SYNC FB EN PGND EP AGND

VCC
TP113 TP1.0
C111 10u/25(X5R)
R108 OPEN
R113 0
R109 750kF
R111 820kF
C113 68p/50(COG)
R110 91kF
R121 143kF
C122 22u/10(X5R)
C112 22u/10(X5R)
FB102 BLM18PG330SN1D
C123 OPEN
R107 27mF(1/4W)
VOUT
TP111 TP1.0
TP112 TP1.0
CN111
ULO-PH409-1*2SF11
Through Hole for
Checking VOUT Voltage

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